CHAPTER 10

CONIC SECTIONS

CHAPTER ESSENCE AT A GLANCE:

Slicing cones creates the 4 iconic geometric curves! Parabola focuses satellite dishes; Ellipse describes planetary orbits; Hyperbola guides navigation systems. All are governed by the distance ratio e = (Distance to Focus) over (Distance to Directrix)!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Circle Equation:** $(x - h)^2 + (y - k)^2 = r^2$ with center (h, k) and radius r. General form: $x^2 + y^2 + 2gx + 2fy + c = 0$, center (-g, -f), r = sqrt($g^2 + f^2 - c$).

• **Parabola $y^2 = 4ax$:** Focus: (a, 0) ; Directrix: x = -a ; Axis: y = 0 ; Length of Latus Rectum = 4a. If opening along y-axis: $x^2 = 4ay$.

• **Ellipse x^2 over a^2 + y^2 over b^2 = 1 (a > b):** Foci: (±c, 0) where $c^2 = a^2 - b^2$; Eccentricity: e = c over a < 1 ; Vertices: (±a, 0) ; Length of Latus Rectum = $2b^2$ over a.

• **Hyperbola x^2 over a^2 - y^2 over b^2 = 1:** Foci: (±c, 0) where $c^2 = a^2 + b^2$; Eccentricity: e = c over a > 1 ; Vertices: (±a, 0) ; Length of Latus Rectum = $2b^2$ over a.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$(x-h)^2+(y-k)^2=r^2 \quad \text{(Circle with Centre } (h,k) \text{ and Radius } r)$$

Cartesian Equation of Circle

$$y^2=4ax \rightarrow \text{Focus: } (a,0), \quad \text{Directrix: } x=-a, \quad \text{Latus Rectum: } 4a$$

Standard Rightward Parabola Canonical Parameters

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \ (a>b) \rightarrow c=\sqrt{a^2-b^2}, \quad e=\frac{c}{a}<1, \quad \text{LR}=\frac{2b^2}{a}$$

Standard Ellipse Foci, Eccentricity, and Latus Rectum

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1 \rightarrow c=\sqrt{a^2+b^2}, \quad e=\frac{c}{a}>1, \quad \text{LR}=\frac{2b^2}{a}$$

Standard Horizontal Hyperbola Fundamental Constants

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In an ellipse, $c^2 = a^2 - b^2$ (minus sign!). In a hyperbola, $c^2 = a^2 + b^2$ (plus sign!). Do not mix them up!

• **Alert:** Eccentricity tells you the curve immediately: Circle (e = 0), Ellipse (e < 1), Parabola (e = 1), Hyperbola (e > 1)!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Find the equation of the circle with centre (0, 2) and radius 2. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Standard Form]** $(x - h)^2 + (y - k)^2 = r^2$. $(x - 0)^2 + (y - 2)^2 = 2^2 \Rightarrow x^2 + y^2 - 4y + 4 = 4 \Rightarrow x^2 + y^2 - 4y = 0$. [2 Marks]

• **[Boxed Result: $x^2 + y^2 - 4y = 0$]**

2. Find the coordinates of the focus, axis of parabola, equation of directrix, and length of latus rectum for $x^2 = -16y$. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Parameters]** Comparing with $x^2 = -4ay$: $4a = 16 \Rightarrow a = 4$. Axis is the y-axis ($x = 0$). Focus = $(0, -a) = (0, -4)$. Directrix: $y = a \Rightarrow y = 4$. Length of latus rectum = $4a = 16$. [2 Marks]

• **[Boxed Result: Focus = (0, -4), Directrix: y = 4, Latus Rectum = 16]**

3. Find the coordinates of the foci, vertices, length of major axis, minor axis, and eccentricity of the ellipse: x^2 over 49 + y^2 over 36 = 1. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Identifying Parameters]** $a^2 = 49 \Rightarrow a = 7$, $b^2 = 36 \Rightarrow b = 6$ (Major axis along x-axis since $a > b$). $c = \sqrt{(a^2 - b^2)} = \sqrt{(49 - 36)} = \sqrt{(13)}$. [1 Mark]

• **[Step 2: Foci and Vertices]** Foci = $(\pm c, 0) = (\pm \sqrt{(13)}, 0)$. Vertices = $(\pm a, 0) = (\pm 7, 0)$. [1 Mark]

• **[Step 3: Axes and Eccentricity]** Length of major axis = $2a = 14$. Length of minor axis = $2b = 12$. Eccentricity e = c over a = $\sqrt{(13)}$ over 7. [1 Mark]

• **[Boxed Result: Foci = $(\pm \sqrt{(13)}, 0)$, Vertices = $(\pm 7, 0)$, e = $\sqrt{(13)}$ over 7]**

4. Find the coordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbola: x^2 over 16 - y^2 over 9 = 1. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Hyperbola Parameters]** $a^2 = 16 \Rightarrow a = 4$, $b^2 = 9 \Rightarrow b = 3$. For hyperbola: $c^2 = a^2 + b^2 = 16 + 9 = 25 \Rightarrow c = 5$. [1 Mark]

• **[Step 2: Foci and Vertices]** Transverse axis along x-axis. Vertices = $(\pm a, 0) = (\pm 4, 0)$. Foci = $(\pm c, 0) = (\pm 5, 0)$. [1 Mark]

• **[Step 3: Eccentricity and Latus Rectum]** Eccentricity e = c over a = 5 over 4 = 1.25. Length of latus rectum = 2 b^2 over a = 2(9) over 4 = 18 over 4 = 9 over 2 = 4.5. [1 Mark]

• **[Boxed Result: Vertices = (± 4, 0), Foci = (± 5, 0), e = 5 over 4, Latus Rectum = 9 over 2]**

5. Find the equation of the circle passing through the points (4, 1) and (6, 5) and whose centre is on the line 4x + y = 16. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Centre Condition]** Let centre be (h, k). Since it lies on $4x + y = 16$: $4h + k = 16 \Rightarrow k = 16 - 4h$ ---(1). [1 Mark]

• **[Step 2: Equal Radii from Centre]** $(h - 4)^2 + (k - 1)^2 = (h - 6)^2 + (k - 5)^2$. $h^2 - 8h + 16 + k^2 - 2k + 1 = h^2 - 12h + 36 + k^2 - 10k + 25$. $-8h - 2k + 17 = -12h - 10k + 61 \Rightarrow 4h + 8k = 44 \Rightarrow h + 2k = 11$ ---(2). [1 Mark]

• **[Step 3: Solve for h, k and Radius]** From (1) and (2): $h + 2(16 - 4h) = 11 \Rightarrow h + 32 - 8h = 11 \Rightarrow -7h = -21 \Rightarrow h = 3$. $k = 16 - 4(3) = 4$. Centre = (3, 4). $r^2 = (3 - 4)^2 + (4 - 1)^2 = (-1)^2 + 3^2 = 1 + 9 = 10$. Equation: $(x - 3)^2 + (y - 4)^2 = 10 \Rightarrow x^2 + y^2 - 6x - 8y + 15 = 0$. [1 Mark]

• **[Boxed Result: $x^2 + y^2 - 6x - 8y + 15 = 0$]**

6. Find the equation of the hyperbola whose foci are (0, ± 12) and the length of the latus rectum is 36. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Parameters Identification]** Foci on y-axis ⇒ standard form: y^2 over a^2 - x^2 over b^2 = 1. Foci (0, ± c) = (0, ± 12) ⇒ c = 12. [1 Mark]

• **[Step 2: Latus Rectum]** Length of latus rectum = 2 b^2 over a = 36 ⇒ b^2 = 18 a. [1 Mark]

• **[Step 3: Hyperbola Relation $c^2 = a^2 + b^2$]** $c^2 = a^2 + b^2$ ⇒ $12^2 = a^2 + 18$ a ⇒ $a^2 + 18$ a - 144 = 0. (a + 24)(a - 6) = 0 ⇒ a = 6 (since a > 0). b^2 = 18(6) = 108. a^2 = 36. [1.5 Marks]

• **[Step 4: Equation]** y^2 over 36 - x^2 over 108 = 1 (or 3 y^2 - x^2 = 108). [0.5 Mark]

• **[Boxed Result: y^2 over 36 - x^2 over 108 = 1]**

7. Find the equation of the circle passing through the points (2, 3) and (-1, 1) and whose centre lies on the line x - 3y - 11 = 0. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Centre on Line]** Let centre be C(h, k). Lies on x - 3y - 11 = 0 ⇒ h - 3k = 11 ⇒ h = 3k + 11 ---(1). [1 Mark]

• **[Step 2: Equal Distance Condition]** $CP^2 = CQ^2$: $(h - 2)^2 + (k - 3)^2 = (h + 1)^2 + (k - 1)^2$. $h^2 - 4h + 4 + k^2 - 6k + 9 = h^2 + 2h + 1 + k^2 - 2k + 1$. -4h - 6k + 13 = 2h - 2k + 2 ⇒ 6h + 4k = 11 ---(2). [1.5 Marks]

• **[Step 3: Solve for h, k and Radius]** Substitute (1) into (2): 6(3k + 11) + 4k = 11 ⇒ 18k + 66 + 4k = 11 ⇒ 22k = -55 ⇒ k = -5 over 2. h = 3(-5 over 2) + 11 = -15 over 2 + 22 over 2 = 7 over 2. Centre = (7 over 2, -5 over 2). r^2 = (7 over 2 - 2)^2 + (-5 over 2 - 3)^2 = (3 over 2)^2 + (-11 over 2)^2 = 9 over 4 + 121 over 4 = 130 over 4 = 65 over 2. [1 Mark]

• **[Step 4: Circle Equation]** (x - 7 over 2)^2 + (y + 5 over 2)^2 = 130 over 4 ⇒ x^2 - 7x + 49 over 4 + y^2 + 5y + 25 over 4 = 130 over 4 ⇒ x^2 + y^2 - 7x + 5y - 14 = 0. [0.5 Mark]

• **[Boxed Result: $x^2 + y^2$ - 7x + 5y - 14 = 0]**